Information, Topology and Evaluation in Sensor Placement for Urban Drainage Networks: A Literature Review

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Abstract. Sewer and stormwater networks are large, mostly buried and expensive to monitor, so a utility can place sensors at only a small share of its manholes. Choosing those few locations is the *optimal sensor placement* (OSP) problem. This review brings together four sources on the problem: one review article, one journal paper on optimising and evaluating a drainage sensor network, one conference paper that uses network topology alone, and one earlier paper that set out an entropy-based criterion for hydrometric networks. The sources are organised by the idea each one contributes: information theory as the basis for judging what a sensor is worth, a full optimise-then-evaluate framework for a drainage network, a topology-only screening index that needs no hydraulic model, and a review that places these approaches side by side. Across the sources, three themes recur: placement should add information without repeating what other sensors already capture, the quality of a sensor layout should be measured and not assumed, and the data a method needs ranges from a pipe map alone to tens of thousands of calibrated simulations. The review closes by identifying five open gaps, most notably the lack of a standard way to evaluate a layout and the absence of any method aimed at capacity that is consumed gradually by new housing, and relates these to a capstone project on sensor placement for growth in an Adelaide sewer network.

1 Introduction

Urban drainage networks carry sewage and stormwater through pipes that are mostly out of sight, which makes their day-to-day condition hard to know. Sensors that measure water level, flow or water quality can show how a network is operating and can warn of blockages, overflows, pollution and flooding (Ma et al., 2025). Each sensor, however, costs money to buy, install and maintain, so only a small number of the possible locations can be used (Zheng et al., 2025). This leads to the question at the centre of this review: given a limited number of sensors, where should they go so that the data they collect is as useful as possible?

This review covers four sources on that question. They differ in age, scope and the amount of data they need, but each adds a distinct idea. They are organised as follows:

1. **Information-theory foundations** (Section 2): a criterion for choosing stations that carry the most information with the least duplication (Li et al., 2012).
2. **Optimisation and evaluation in a drainage network** (Section 3): a framework that selects sensor locations with information measures and then tests the chosen layout by reconstructing water levels at the unmonitored nodes (Zheng et al., 2025).

3. **Topology-only analysis** (Section 4): an index of how easily a spill at each node would be detected at the outfall, worked out from the network map alone (Simone et al., 2022).
4. **Review of placement strategies** (Section 5): a structured review of sensor placement in urban drainage networks and a general framework for it (Ma et al., 2025).

Section 6 draws out common themes, compares how each source treats what a sensor can observe, sets out the methods side by side and lists open gaps. Section 7 relates these to the capstone project, and Section 8 concludes.

2 Information-Theory Foundations

Information theory treats a sensor reading as a reduction in uncertainty. A location whose readings vary a great deal has high entropy and so can tell an operator more, while two locations whose readings move together largely repeat each other. Several of the later methods in this review build on these ideas.

2.1 Entropy measures used for network design

Li et al. (2012) work with four measures. The marginal entropy of a station's record X is

$$H(X) = -\sum_{k} p(x_k) \log p(x_k), \tag{1}$$

the information that station holds on its own. The joint entropy $H(X_1, \ldots, X_N)$ is the information a group of stations holds together. Transinformation (mutual information),

$$T(X_1, X_2) = H(X_1) + H(X_2) - H(X_1, X_2), \tag{2}$$

is the information two stations share, and so how much of one can be inferred from the other. Total correlation,

$$C(X_1, \ldots, X_N) = \sum_{i=1}^{N} H(X_i) - H(X_1, \ldots, X_N), \tag{3}$$

is the duplicated information within a group, which is a measure of redundancy.

2.2 The maximum information minimum redundancy criterion

Earlier entropy-based designs usually captured only one or two of these properties, and some assumed that the data were normally distributed (Li et al., 2012). Li et al. (2012) propose a single criterion, *maximum information minimum redundancy* (MIMR), which asks a selected set of stations S to hold as much information as possible, to share as much information as possible with the stations left out (F), and to repeat as little as possible among themselves. The three aims are combined into one objective with trade-off weights $\lambda_1 + \lambda_2 = 1$:

$$\max \; \lambda_1 \Big(H(X_{S_1}, \ldots, X_{S_k}) + \sum_{i=1}^{m} T\big(\langle X_{S_1}, \ldots, X_{S_k} \rangle; X_{F_i}\big) \Big) - \lambda_2 \, C(X_{S_1}, \ldots, X_{S_k}). \tag{4}$$

Stations are chosen one at a time by a greedy ranking: at each step every remaining candidate is tested and the one that most improves Eq. (4) is added. Selection can stop once the chosen set explains a set share of the total information, for example 90%.

To avoid assuming a distribution, continuous records are first converted to discrete values with a floor function, $x_q = a\lfloor(2x + a)/(2a)\rfloor$. The bin width a can carry physical meaning. In one of the case studies it was set to 5 cm, because smaller changes in water level were attributed to wind, boats or pump operation and treated as noise (Li et al., 2012).

2.3 Case studies and findings

The criterion was tested on two networks. The first was 12 stream gauges on the main stem of the Brazos River in Texas, where the existing network was evaluated. The second selected monitors from 1,520 candidate points along the canals of a polder system in Delfland, the Netherlands, using water levels generated by the water board's hydrodynamic model for a storm event (Li et al., 2012). Compared with an earlier entropy-based approach by Alfonso and colleagues, MIMR captured a similar share of total information on the river (about 80% for the top six stations against 82%) and more on the polder, while choosing stations that were more widely spread and contained less duplicated information. On the polder the top ten MIMR monitors were spread across the canal system without clusters, whereas the comparison method produced clusters of nearby stations (Li et al., 2012).

The authors note three limitations. The greedy search is slower than the matrix-based comparison method, because every remaining candidate is scanned at each step. Cost is not included, although a penalty in information units could be added. Most importantly for later work, MIMR can only choose stations from an existing dense set of candidates; it cannot by itself say how to extend a sparse network. The authors suggest that a physically based model can generate a synthetic dense network for this purpose, as was done in the polder case, provided the synthetic data keep the properties of the real system (Li et al., 2012).

3 Optimisation and Evaluation of a Drainage Sensor Network

Zheng et al. (2025) start from the observation that most earlier work on drainage sensor placement chose locations but did not test how well the chosen network then performed. Their framework does both: it selects locations with information measures and then evaluates the result by how well the sensors' readings can rebuild conditions at the nodes that are not monitored.

3.1 Scenarios from a calibrated model

The study area is a 2.679 km^2 residential catchment in HZ city, China, modelled in SWMM with 878 nodes, 898 conduits and 2,693 subcatchments. Three water-level sensors were already in place. The model was calibrated against one of them (Nash–Sutcliffe efficiency 0.907) and validated against the other two (0.948 and 0.901). To represent uncertainty, 144 design rainfall events (six return periods,

eight durations and three peak positions) were combined with 300 sets of model parameters drawn by Monte Carlo sampling, giving 43,200 simulations. Eighty per cent of these were used to estimate the probabilities behind the information measures and the remaining 20% for evaluation (Zheng et al., 2025).

3.2 Information measures and multi-objective search

Two measures drive the placement. The *value of information* (VOI) of a node expresses how much a reading there helps decide whether other nodes are flooding, summed over the whole network. Transinformation (TE) between selected nodes measures their redundancy, in the same sense as in Li et al. (2012). A multi-objective genetic algorithm (NSGA-II) then searches for layouts that maximise total VOI, minimise total TE and minimise the number of sensors, limited here to at most 20 (Zheng et al., 2025).

Two results about the measures are worth noting. First, VOI was highest when the node being judged had a flooding probability of about 0.5. Nodes that almost never flood, or almost always flood, are predictable, so a sensor adds little about them. Second, in this flood-monitoring task neither VOI nor TE depended on the distance between nodes, unlike water-quality monitoring where nearby sensors tend to give more information and more redundancy. The authors conclude that what makes a good layout depends on the monitoring task (Zheng et al., 2025).

3.3 Evaluation by matrix completion

To test a layout, water levels at all nodes are arranged as a matrix with only the sensor columns filled. Non-negative matrix factorisation approximates this matrix as the product of two smaller non-negative matrices, fitting only the known entries, and so fills in the missing ones. The fitted levels are then compared with the simulated levels, using error measures when there is no flooding and accuracy in classifying flooded nodes during a serious flood (Zheng et al., 2025).

3.4 Findings and limitations

With the same number of sensors (three), the optimised layout had a VOI 0.138 higher and a TE 0.0039 lower than the existing sensors. Optimised layouts also outperformed randomly chosen layouts of the same size. With four sensors the error in reconstructing operating status was 33% and flood-risk accuracy was 76%; with eight sensors these became 29% and 82%. On this basis the authors recommend four sensors for a lower budget and eight for a higher one. The stated limitations are that water level was the only monitored variable, that only one catchment was tested, and that the matrix completion used no prior knowledge such as the network's topology, which the authors expect would improve it (Zheng et al., 2025).

4 Topology-Only Analysis

Hydraulic and water-quality models are effective but need detailed data and heavy computation, which may not be practical for large networks. Simone et al. (2022) instead model the spread of a contaminant using only the network's layout, drawing

on complex network theory and a backtracking algorithm, with no hydraulic or water-quality simulation and no optimisation.

4.1 Spread rules based on Horton order

The network is a directed graph in which pipes are links pointing in the flow direction and manholes are nodes. A contaminant injected at a node travels only downstream, and each node it reaches becomes a new source. In place of flow rates, which would need a hydraulic model, the method uses each pipe's Horton order, a ranking of its position in the branching hierarchy. For the pipe leaving the source node k,

$$C^c_{(k,m)} = C_i \frac{Q_i}{Q_i + \sum_l H_d}, \tag{5}$$

where C_i is the injected concentration, Q_i is the injected flow, set to the lowest Horton order ($Q_i = H_i = 1$), and H_d is the Horton order of the pipes leaving the source. Where pipes join, the downstream concentration is a Horton-order-weighted average of the incoming pipes,

$$C^c_{(n,p)} = \frac{\sum_l H_u C^c_u}{\sum_l H_d}, \tag{6}$$

so an uncontaminated incoming pipe dilutes the flow. Because Horton order changes only where a junction raises it, dilution occurs only at those junctions (Simone et al., 2022).

4.2 The contamination index

Decay along the way is represented by path length instead of time. For each source node s, the contamination index sums the contaminant in each pipe l on the path to the outfall, scaled by a length term,

$$I_{C,s} = \sum_l C^C_l \, \exp(L_l \, f), \tag{7}$$

where L_l is the pipe length and f is a scale factor. A high I_C means that a spill at that node would be easy to detect at the outfall; a low value means it would be hard to detect because of dilution and decay along the way (Simone et al., 2022).

4.3 Case study and limitations

The method was applied to SWMM Example 3, a small network of 32 nodes, 32 pipes, one outfall, one storage unit and one pump. With a spill of 0.1 mg/L injected at each node in turn, I_C was highest for nodes near the outfall and fell for nodes further upstream, which the authors note is consistent with the physics of the problem. They describe the result as a low-cost preliminary analysis that could be applied even where hydraulic behaviour is uncertain (Simone et al., 2022). The paper does not compare I_C with a hydraulic or water-quality simulation, and testing on a real, complex network is left to future work.

5 Review of Placement Strategies

Ma et al. (2025) review how Internet of Things (IoT) sensors are placed in urban drainage networks. A Web of Science search returned 189 papers, of which 96 were kept, supplemented by work on related networks such as water supply and rivers.

5.1 A general framework

The review proposes a framework in three stages: classify the monitoring objective, select a placement method, and evaluate the resulting scheme quantitatively. Objectives fall into two groups. The first is the operating status of the network, monitored through flow, level and velocity, which also supports model calibration. The second is abnormal states: tracing pollutants and illegal discharges (including wastewater-based epidemiology) and monitoring flood and overflow hotspots (Ma et al., 2025).

5.2 Five bases for placement

Placement methods are grouped by the theory behind them (Ma et al., 2025):

- **Engineering experience**: rules such as uniformity, representativeness and access, used especially where old networks lack data for a reliable model, but hard to apply on large, complex networks.
- **Information theory**: entropy-based measures, including the MIMR criterion of Li et al. (2012) and value-of-information methods. A stated drawback is that low-redundancy criteria tend to push sensors as far apart as possible, towards the edges of the study area, which can ignore practical needs.
- **Statistical theory**: mainly cluster analysis, which groups similar nodes. The number of clusters must be chosen in advance, the method does not say where in a cluster to place the sensor, and the value of individual locations is overlooked.
- **Complex network theory**: topology-based indices that avoid hydraulic simulation, including the work of Simone and colleagues (e.g. Simone et al., 2022).
- **Observability theory**: using the fewest sensors to estimate the state of unmonitored areas. The review finds few such studies for drainage networks and cites the matrix-completion approach of Zheng et al. (2025) as one.

5.3 Evaluation and recommendations

Evaluation measures found in the literature fall into three groups: algorithm performance (for example running time and clustering indices), errors between predicted and actual values (for example RMSE and Nash–Sutcliffe efficiency), and how far monitoring objectives are met (for example detection rate, response time and reliability). The authors conclude that there is still no unified process for evaluating a layout, that some studies do not evaluate their layout at all, and that current

layouts may not be optimal. They recommend hybrid strategies that combine theories, and a phased approach in which sensors are first placed at critical nodes and later stages are adjusted using the data and information gained (Ma et al., 2025).

6 Synthesis and Research Gaps

6.1 Cross-cutting themes

- **A shared line of ideas.** The four sources are closely linked. Ma et al. (2025) cite each of the other three: MIMR as the main information-theory criterion, Zheng et al. (2025) as an example of observability through reconstruction, and Simone et al. (2022) as a topology-based alternative. Zheng et al. (2025) in turn rely on transinformation in the same sense as Li et al. (2012).
- **Information without repetition.** Li et al. (2012) and Zheng et al. (2025) both treat redundancy as something to be minimised alongside information, and Ma et al. (2025) restate this as the aim of placement in general. Selecting by information content alone is shown by Li et al. (2012) to produce clusters of stations that repeat each other.
- **Evaluation as a separate step.** Zheng et al. (2025) and Ma et al. (2025) both argue that a layout should be tested after it is chosen, and both note that this step is often missing.
- **A wide range in data needs.** The methods range from a pipe map with flow directions and lengths (Simone et al., 2022), through long records at many candidate points (Li et al., 2012), to tens of thousands of runs of a calibrated model (Zheng et al., 2025).

6.2 What each source assumes a sensor can see

The sources differ in how they model what a sensor at a given location can observe. In Li et al. (2012) and Zheng et al. (2025) a sensor reads its own location, and its usefulness comes from the statistical relationship between its record and the records elsewhere. Neither applies a separate model of measurement error, although Li et al. (2012) use the bin width a to ignore changes smaller than a physical noise level. In Simone et al. (2022) the only observation point is the outfall, and I_C describes how visible each upstream node is from there; this makes detectability depend directly on position in the network, through dilution at junctions and length along the path. Ma et al. (2025) do not adopt a single model but note that topology-based methods can overlook hydraulic detail and that information-based methods can be sensitive to noise. None of the four sources varies a sensor's own accuracy with location or pipe size.

6.3 Comparison of methods

Table 1 summarises the sources by method, objective, data needed and scale of the case study.

Table 1: Comparison of the reviewed sources.

Source	Primary method	Monitoring objective	Data needed	Case-study scale
Li et al. (2012)	Entropy criterion (MIMR) with greedy ranking	General hydrometric network design	Records at a dense set of candidate stations	12 river gauges; 1,520 polder points (modelled)
Zheng et al. (2025)	VOI and transinformation with NSGA-II; matrix completion for evaluation	Flood risk and operating status	Calibrated SWMM model; 43,200 simulations	878 nodes, one catchment
Simone et al. (2022)	Horton-order spread rules and contamination index	Detectability of contaminant spills	Network map, flow directions, pipe lengths	32 nodes (SWMM Example 3)
Ma et al. (2025)	Structured review and general framework	Status and abnormal states	Not applicable	96 studies

6.4 Research gaps

1. **No standard way to evaluate a layout.** Studies judge layouts by different measures, and some do not evaluate them at all (Ma et al., 2025). Zheng et al. (2025) offer one method, reconstruction by matrix completion, but it has been shown on a single catchment.

2. **Testing on real, large networks.** Simone et al. (2022) use a 32-node example network and leave real networks to future work; Zheng et al. (2025) use one catchment; Li et al. (2012) work outside urban drainage.

3. **Extending sparse networks.** MIMR can only choose from an existing dense set of candidates, and the suggested remedy, a synthetic dense network from a model, depends on that model representing the real system (Li et al., 2012).

4. **Cost in the objective.** Li et al. (2012) leave cost out and note that expressing it in information units is unresolved; Zheng et al. (2025) use the number of sensors as a stand-in for cost.

5. **Phased and adaptive placement.** All three primary studies produce a single, fixed layout. Ma et al. (2025) identify phased deployment with feedback as a direction that remains largely unexplored.

7 Relevance to the Capstone Project

The capstone behind this review asks where to place sensors in an Adelaide sewer network so that the effect of new housing on pipe capacity can be seen over time.

None of the four sources addresses this directly: they deal with flooding, contaminant spills and general hydrometric design, all under the network's current conditions rather than a load that grows gradually. Each still offers something that can be carried across.

- **Synthetic candidates are acceptable, with care.** The project has no measured levels, so its candidate data come from a SWMM model. Li et al. (2012) support this use of model-generated dense networks, with the caution that the model must preserve the properties of the real system. This makes calibration a condition of trusting any information-based placement, not an optional step.
- **Evaluate by reconstruction.** Following Zheng et al. (2025), a placement can be judged by how well its readings rebuild conditions at unmonitored manholes, compared with random layouts of the same size, and the number of sensors can be justified by where the benefit levels off.
- **Avoid redundant sites.** The finding that selecting by information alone produces clusters of near-duplicate stations (Li et al., 2012) is a reason to check any proposed set of manholes for overlap, for example sites on the same branch that see the same change.
- **Screen with the pipe map first.** A topology-only index in the spirit of Simone et al. (2022) could be computed from the published pipe network before any hydraulic runs, to show which manholes are well placed to see changes upstream. This would need its own check against the hydraulic model, which the source does not provide.
- **Plan in stages.** Because housing arrives over years, the phased deployment and feedback suggested by Ma et al. (2025) matches the problem: start at critical nodes and revise later sensors using the data collected.

8 Conclusion

The four sources reviewed here show a clear line of development, from a general entropy criterion for hydrometric networks (Li et al., 2012), through a full optimise-and-evaluate framework for a drainage catchment (Zheng et al., 2025) and a topology-only screening index (Simone et al., 2022), to a review that places these approaches in one framework (Ma et al., 2025). Together they suggest that a sensor layout should add information without repeating it, that its quality should be measured after it is chosen rather than assumed, and that the right method depends on the monitoring task and on the data available. They also leave open how to evaluate layouts in a standard way, how to test methods on large real networks, and how to place sensors in stages as conditions change. For a capstone project on sensor placement under housing growth, the most useful lessons are to evaluate any placement by reconstruction against simple baselines, to check selected sites for redundancy, and to treat a calibrated model as the condition on which an information-based placement can be trusted.

References

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